

Data Hiding using Steganography

Tool: Steghide

Theory

Steganography is the technique of hiding secret information inside an ordinary file such as an image or audio file so that its existence is not visible to anyone other than the intended recipient. Unlike encryption which makes data unreadable, steganography hides the fact that a secret message exists at all. Steghide is a command-line tool that embeds and extracts hidden data inside JPEG and BMP image files using a passphrase for protection.

Steps

1. Open the terminal in Kali Linux. Install Steghide. Run: `sudo apt install steghide -y`
 2. Navigate to the Desktop. Run: `cd ~/Desktop` — Download a sample JPEG image to use as the cover file. Run: `wget -O cover.jpg https://samples.exiftool.org/Canon.jpg`
 3. Create a secret message file to hide inside the image. Run: `echo 'This is a secret message' > secret.txt` — Verify it was created. Run: `cat secret.txt`
 4. Embed the secret file inside the cover image. Run: `steghide embed -cf cover.jpg -sf secret.txt` — When prompted, enter a passphrase such as `mypass123` and confirm it. The tool outputs: `embedding secret.txt in cover.jpg... done`
 5. Verify the image looks unchanged. Open `cover.jpg` in an image viewer. The image appears identical to before — the hidden data is not visible.
 6. Check information about what is hidden inside the image. Run: `steghide info cover.jpg` — Enter the passphrase when prompted. The output shows the embedded filename and its size.
 7. Delete the original `secret.txt` to simulate the receiver's side. Run: `rm secret.txt` — Then extract the hidden file from the image. Run: `steghide extract -sf cover.jpg` — Enter the passphrase when prompted.
 8. Verify the extracted file contents. Run: `cat secret.txt` — The original secret message should appear, confirming successful extraction.
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Conclusion

Steghide successfully hid a secret text file inside a JPEG image and extracted it without any visible change to the cover image. This demonstrates how steganography can be used to conceal the existence of sensitive data within ordinary media files.